For Performance Measurement

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY 9189/3

PAPER 3 Multiple Choice

NOVEMBER 2017 SESSION 1 hour

Additional materials:
- Data Booklet
- Mathematical tables and/or calculator
- Multiple Choice answer sheet
- Soft clean eraser
- Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 14 printed pages and 2 blank pages.

Section A

*For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the one you consider to be correct.*

1 Which equation relates to the first ionisation energy of chlorine?

A $Cl_{2(g)} \rightarrow Cl^{+}_{2(g)} + e^{-}$

B $Cl_{(g)} \rightarrow Cl^{+}_{(g)} + e^{-}$

C $\frac{1}{2}Cl_{2(g)} \rightarrow Cl^{-}_{(g)} + e^{-}$

D $\frac{1}{2}Cl_{2(g)} \rightarrow Cl^{+}_{(g)} + e^{-}$

2 The combustion of 4 moles of an element, Z, required 89.6 dm^3 of oxygen gas at s.t.p.

What is the empirical formula of the oxide formed?

A ZO
B ZO_2
C Z_4O
D Z_2O

3 Which element exists as discrete molecules in the solid state at room temperature?

A aluminium
B iodine
C argon
D carbon dioxide

4 Which gas is likely to deviate the most from ideal gas behaviour?

A N_2
B CO_2
C O_2
D HCl

5 A solid melts at 114 °C and its boiling point is 184 °C. The solid is insoluble in water but soluble in benzene. The solid does not conduct electricity when solid or when molten.

What structure is the solid most likely to have?

A ionic
B metallic
C simple molecular
D giant molecular

6 An element, X, forms both a dichloride, XCl_2 and a tetrachloride, XCl_4. 10.00 g of the dichloride reacts with excess chlorine to form 12.55 g of the tetrachloride.

The atomic mass of X is

A 73
B 119
C 125
D 207

7 The rate of a particular reaction doubles for each 10 °C temperature rise. The reaction takes 200 seconds to be completed at 30 °C.

How long will the reaction take to be completed at 70 °C?

A 12.5 seconds
B 100.0 seconds
C 466.7 seconds
D 1 600.0 seconds

8 Which species is the most readily reduced?

A H_2O

B H_2O_2

C $S_4O_6^{2-}$

D $S_2O_8^{2-}$

9 A buffer solution was made by dissolving 1.5 g of sodium ethanoate, $CH_3COO^-Na^+$, in 1 dm^3 of 0.100 mol/dm^3 ethanoic acid CH_3COOH.

$\left[K_a = 1.75 \times 10^{-5} \text{ mol/dm}^3\right]$

What was the pH of the buffer solution?

A 3.58
B 4.02
C 5.49
D 5.93

10 The dissolution of an ionic solid is described by

$$AB_{(s)} \rightleftharpoons A^+_{(aq)} + B^-_{(aq)}.$$

If the solubility of AB is 0.0427 mol/dm^3, the K_{sp} of AB is

A $1.82 \times 10^2 mol^2/dm^6$.
B 2.14×10^{-2} mol^2/dm^6.
C 1.82×10^{-3} mol^2/dm^6.
D 8.50×10^{-3} mol^2/dm^6.

11 Which forward reaction is more favoured by an increase in pressure?

A $P_{4(s)} + 6Cl_{2(g)} \rightleftharpoons 4PCl_{3(g)}$

B $PCl_{3(g)} + 3NH_{3(g)} \rightleftharpoons P(NH_2)_{3(g)} + 3HCl_{(g)}$

C $4HCl_{(g)} + O_{2(g)} \rightleftharpoons 2H_2O_{(g)} + 2Cl_{2(g)}$

D $2H_2O_{2(l)} \rightleftharpoons 2H_2O_{(l)} + O_{2(g)}$

12 Which conditions of pressure and temperature would give the most accurate value of the relative molecular mass, M_r, of a gaseous compound?

	pressure	temperature
A	high	high
B	high	low
C	low	high
D	low	low

13 The solubilities of Group (II) sulphates decrease as proton number of the metal increases.

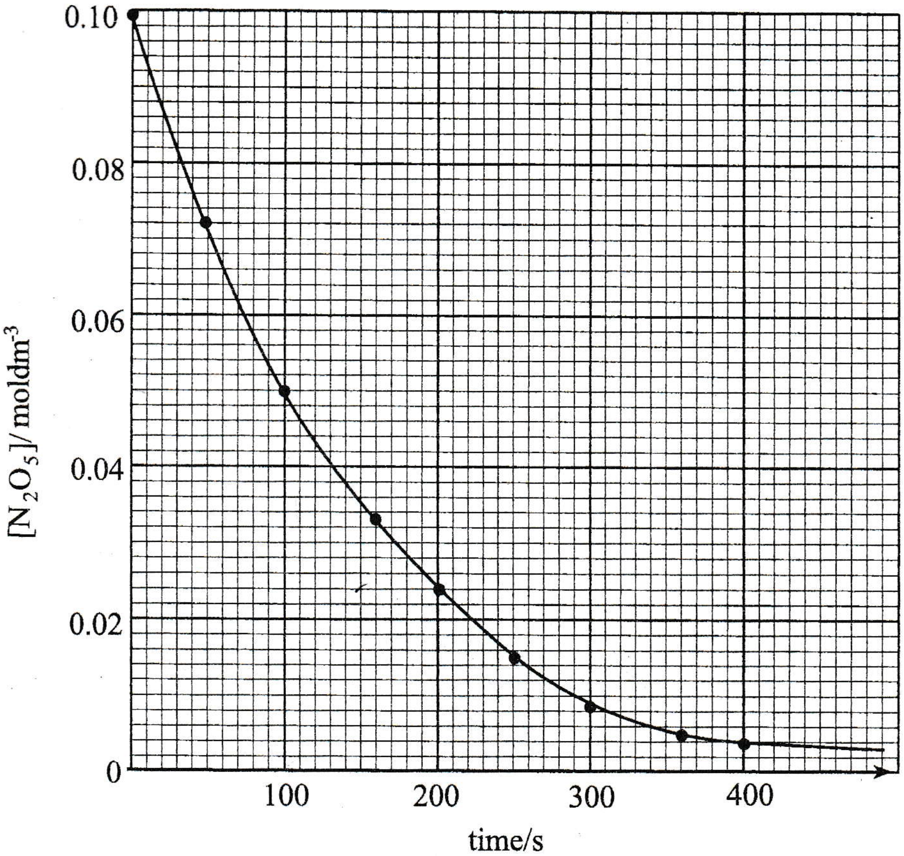
Which factor causes this trend?

A the ionic radius of the metal ion
B the enthalpy change of formation of the sulphate
C the enthalpy change of hydration of the metal ion
D the ionic charge of the metal ion

14 $N_2O_{5(g)}$ decomposes to $NO_{2(g)}$ and $O_{2(g)}$ according to the equation:

$$2N_2O_{5(g)} \rightarrow 4NO_{2(g)} + O_{2(g)}$$

The graph shows how the concentration of $N_2O_{5(g)}$ varied with time during the decomposition.



The half-life, for the reaction is

A 100 s.
B 200 s.
C 50 s.
D 400 s.

15 Which statement about the properties of the tetrachlorides of Group (IV) is correct?

A Their melting points decrease down the group.
B Their melting points increase down the group.
C CCl_4 is the only non-polar tetrachloride.
D Tin (IV) chloride is the only ionic tetrachloride.

16 When a solution containing 2.734 g of a complex soluble salt of $CaCl_3.6H_2O$ is treated with excess $AgNO_3$, 2.87 g of $AgCl$ is precipitated.

What is the total number of chloride ions present in each mole of the complex salt?

A 1
B 2
C 3
D 5

17 The metal which has an unpaired electron in the 4s orbital in its ground state is

A chromium.
B cobalt.
C iron.
D manganese.

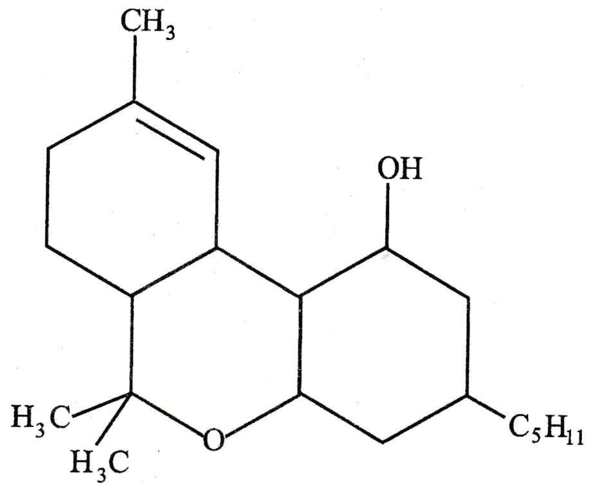
18 Which substance can react with nitrogen monoxide to form non-poisonous products?

A carbon dioxide
B carbon monoxide
C oxygen
D lead-tetraethyl

19 Which statement describes how sulphur dioxide prevents spoilage of food by microbial activity?

A oxidises bacteria cell walls thus destroying them
B produces a choking pungent smell toxic to bacteria
C increases the pH of food
D lowers the pH of food

20 The diagram shows the structure of an organic compound.



How many chiral centres does the compound have?

A 7
B 6
C 5
D 4

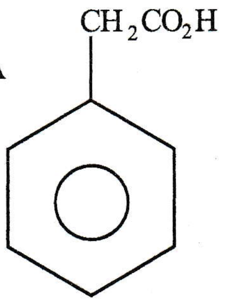
21 Which compound exhibits cis-trans isomerism?

A $C_2H_6O_2$
B $C_2H_2O_4$
C C_2H_3Br
D $C_2H_2Br_2$

22 Ethylbenzene was heated under reflux with alkaline manganese (VII) solution. The resulting mixture was then acidified.

Which formula correctly represents the structure of the organic compound formed?

A



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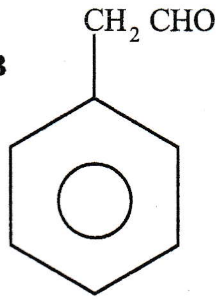
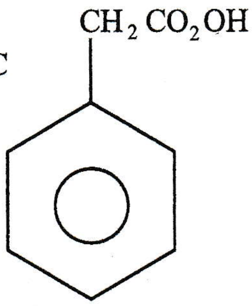
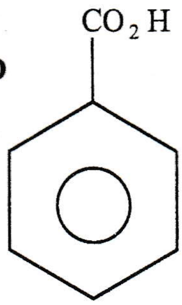


C



D



23 Compound Z was refluxed in ethanolic potassium cynide. The intermediate formed was then hydrolysed by aqueous hydrochloric acid. The final product formed was butanoic acid.

Compound Z could be

A $CH_3CH_2CH_2Br$.
B $CH_3CH_2CH_2CH_2Br$.
C $CH_3CH(OH)CH_2OH$.
D CH_3CH_2CHO.

24 The iodoform reaction can be used to test for the presence of methylketones.

Which species is the oxidising agent in this reaction?

A $OH^-_{(aq)}$

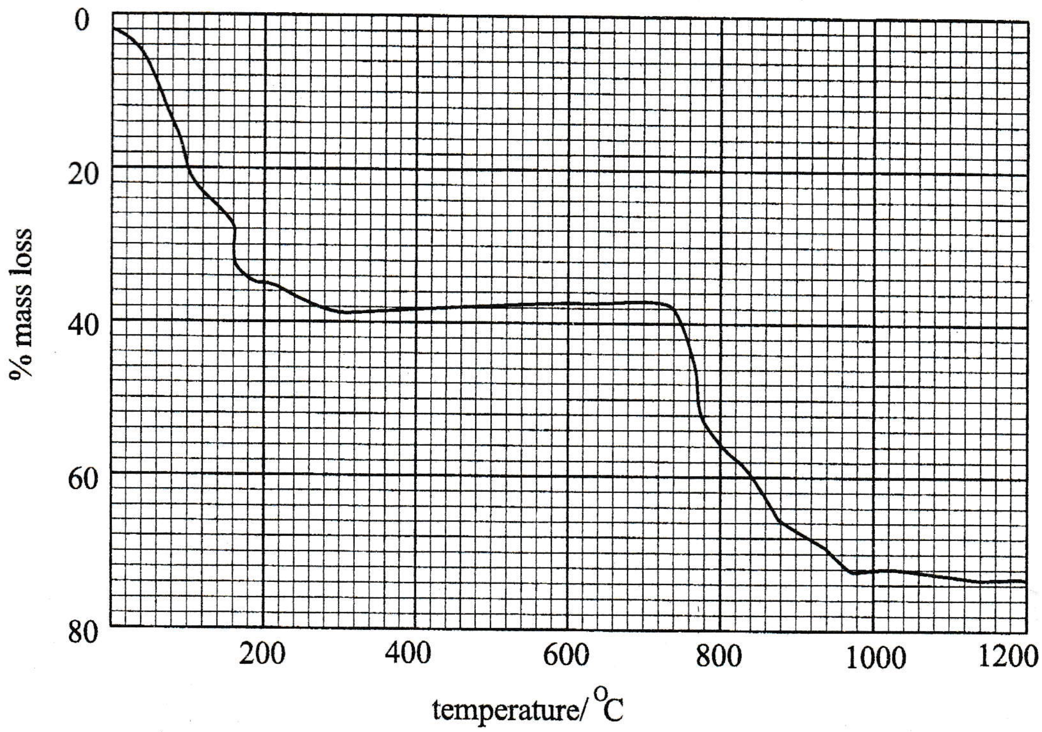
B $I_{2(aq)}$

C $IO^-_{(aq)}$

D $IO^-_{4(aq)}$

25 The amount of water of crystallisation, x, in a compound, $XMO_4.xH_2O$, can be determined by heating the compound and measuring the change in mass.

The graph shows the results of such an investigation.

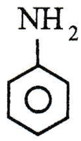


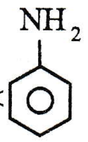
The percentage of water of crystallisation in the compound is approximately

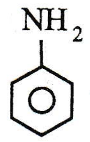
A 10.
B 20.
C 30.
D 40.

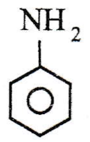
[Turn over

26 Which is the correct order of increasing basicity of phenylamine, ammonia and ethylamine?

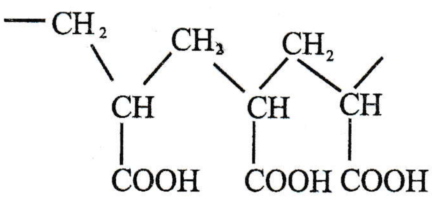
A  $< NH_3 < CH_3\,CH_2\,NH_2$

B $CH_3\,CH_2\,NH_2 < NH_3 <$ 

C $CH_3CH_2\,NH_2 <$  $< NH_3$

D $NH_3 < CH_3\,CH_2NH_2 <$ 

27 The structural formula of a polymer is shown.



From which monomer could the polymer be obtained?

A $Cl\ CH_2CH_2CO_2H$

B $HO_2CCHCH\ CO_2H$

C $H_2C\ CH\ CO_2H$

D $CH_2CHCOOH$

28 Phenylamine is treated with aqueous hydrochloric acid and the solution is then evaporated.

What is the formula of the product?

A $C_6H_5N_2^+Cl^-$

B $C_6H_5NH_4^+Cl^-$

C $C_6H_5NH_3^+Cl^-$

D $C_6H_5NH^+Cl^-$

29 One mole of an organic compound, X, reacts with sodium to give one mole of hydrogen gas.

What could be compound X?

A $CH_3CH_2CH_2CH_2OH$
B $(CH_3)_3OH$
C $CH_3CH_2CH_2CO_2H$
D $CH_3CH(OH)CO_2H$

30 The hydrocarbon, 2,2-dimethylbutane, reacts with chlorine in the presence of u.v light.

Which statement is **not** true about this reaction?

A different forms of chlorohydrocarbons are formed
B the reaction is a free radical substitution reaction
C free radicals are formed in the initiation step only
D free radicals only react in the termination step

Section B

*For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.*

Decide whether each of the statements is or is not correct. (You may find it helpful to put a tick against the statement(s) which you consider to be correct).

*The responses **A** to **D** should be selected on the basis of*

A	B	C	D
1, 2 and 3 are correct	*1 and 2 only are correct*	*2 and 3 only are correct*	*1 only is correct*

No other combination of statements is used as a correct response.

31 Which statement(s) regarding the electronic orbitals of principal quantum numbers 1, 2 and 3 is/are correct?

1. A p-orbital has higher energy than an s-orbital of the same principal quantum number.

2. Each p-orbital can contain a maximum of six electrons.

3. The three 3p orbitals have slightly different energy levels.

32 Which statement(s) is/are true about trends down the Group (VII) hydrides?

1. They become less stable to heat.

2. They become stronger acids.

3. The bond strength increases.

33 Which statement(s) about the commercial extraction of aluminium is/are correct?

1. The anodic reaction is $2O^{2-}_{(l)} \rightarrow O_{2(g)} + 4e^-$.

2. The lining of the electrolytic cell acts as the cathode.

3. The electrolyte is purified Al_2O_3 dissolved in Na_3AlF_6.

34 What can be deduced from the reaction

$$2MO_3^- + SO_3^{2-} + 6H^+ \rightarrow 2MO^{2+} + SO_4^{2-} + 3H_2O?$$

1. MO_3^- ions are the oxidising agent.

2. The number of electrons transferred from each MO_3^- ion is two.

3. The oxidation state of M increases.

35 Hydrogen peroxide reacted with acidified iodide ions, liberating iodine. The results of this reaction, are shown in the table.

initial concentrations of reactants/mol/dm^3			**initial rate of formation of iodine/mol/dm^3/s**
$[H_2O_2]$	$[I^-]$	$[H^+]$	
0.010	0.010	0.10	2×10^{-6}
0.030	0.010	0.10	6×10^{-6}
0.030	0.020	0.10	1.2×10^{-5}
0.030	0.020	0.20	1.2×10^{-5}

What can be deduced from the results?

1. Rate $= k[H_2O_2][I^-]$.

2. The rate constant is $2 \times 10^{-2} mol^{-2} dm^6 s^{-1}$.

3. The half-life of the overall reaction is constant.

36 Silicon carbide has a structure similar to that of diamond.

What is/are the advantages of using a silicon carbide ceramic?

1. Silicon carbide has a high melting point.

2. Silicon carbide is resistant to oxidation.

3. Silicon carbide is less likely to deform under compression.

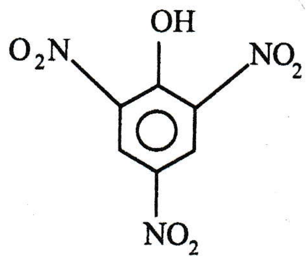
37 Which species can be a ligand in a complex ion?

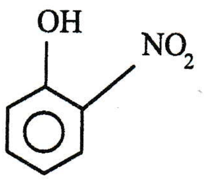
1. NH_4^+

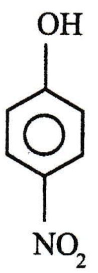
2. CH_3NH_2

3. OH^-

38 Phenol reacts with dilute nitric acid at room temperature to produce

1 

2 

3 

39 The structure of mevalonic acid is shown:

$$HO_2CCH_2\overset{\displaystyle OH \atop |}{C}(CH_3)CH_2CH_2OH$$

Which statement(s) is/are true about mevalonic acid?

1. It has only one chiral carbon atom.
2. It can be esterified by both ethanoic and ethanol.
3. It contains both primary and secondary alcohol groups.

40 What changes are seen when an aqueous solution of bromine is added to a hot dilute aqueous solution of phenylamine?

1. A white precipitate is produced.
2. Bromine is decolourised.
3. Fumes of hydrogen bromide gas are produced.